

Sort an array of 0's, 1's and 2's Given an array nums consisting of only 0, 1 or 2.
Sort the array in non-decreasing order. The sorting must be done in-place, without making a copy of the original array.

I/p: nums = [1, 0, 2, 1, 0]
O/p: [0, 0, 1, 1, 2]

Brute Force

1. Take 3 counters for 0s, 1s and 2s.
2. Loop through the array once and count each number.
3. After this, fill the array based on the frequency of each.

TC: $O(N)$

SC: $O(1)$

Optimized

1. Take low, mid, high
 ↓ ↓ ↓
 0s 1s 2s

2. Find mid

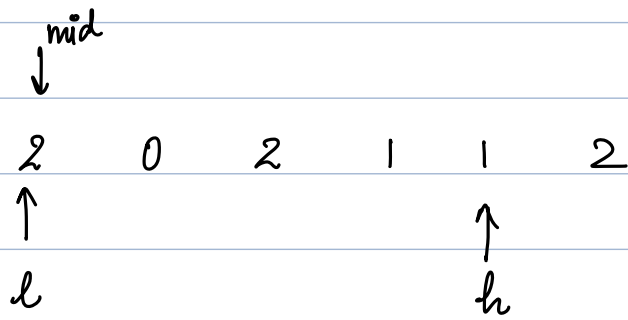
if $arr[mid] == 0$; swap with low, move low and mid

if $arr[mid] == 1$; just move mid

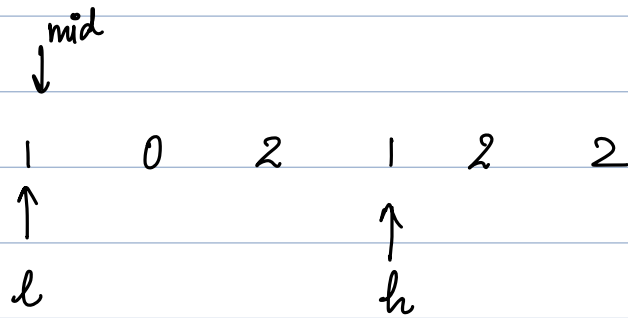
if $arr[mid] == 2$; swap with high, just move high

mid
↓
2 0 2 1 1 2
↑ ↑
l h

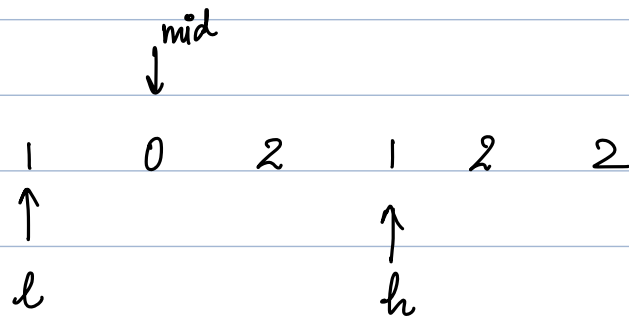
$arr[mid] == 2$, swap with high, just move high



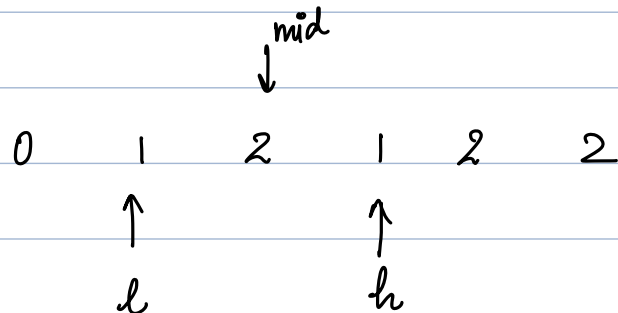
$arr[mid] == 2$, swap with high, just move high



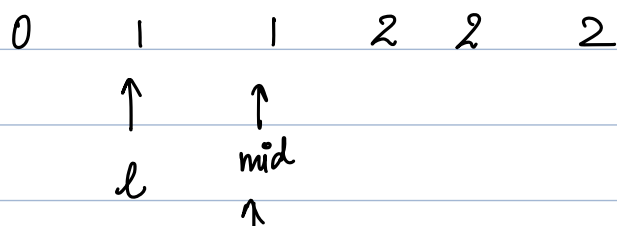
if $arr[mid] == 1$; just move mid



if $arr[mid] == 0$; swap with low, move low and mid



$arr[mid] == 2$, swap with high, just move high



l
h

if $arr[mid] == 1$; just move mid

0	1	1	2	2	2
	↑	↑	↑		
	l	h	mid		

TC: $O(N)$

SC: $O(1)$

```
public void sortZeroOneTwo(int[] nums) {
```

```
    // Initialize three pointers: low and mid at 0, high at the end
```

```
    int low = 0, mid = 0, high = nums.length - 1;
```

```
    // Continue processing until mid crosses high
```

```
    while (mid <= high) {
```

```
        // If current element is 0, swap with low and move both low and mid forward
```

```
        if (nums[mid] == 0) {
```

```
            int temp = nums[low];
```

```
            nums[low] = nums[mid];
```

```
            nums[mid] = temp;
```

```
            low++;
```

```
            mid++;
```

```
        }
```

```
        // If current element is 1, just move mid forward
```

```
        else if (nums[mid] == 1) {
```

```
            mid++;
```

```
        }
```

```
        // If current element is 2, swap with high and move only high backward
```

```
        else {
```

```
int temp = nums[mid];
```

```
nums[mid] = nums[high];
```

```
nums[high] = temp;
```

```
high--;
```

```
}
```

```
}
```

```
}
```